

Ansh Sharma

Backend Engineer FastAPI – Distributed Systems – PostgreSQL – Redis – AWS – Python

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Built and deployed **LinkedOut**, a multi-tenant AI platform on AWS; independently traced and fixed a live cross-tenant data leak and reduced a silent 30% IP-ban rate with no existing observability or debug trail. Strong focus on distributed APIs, async systems, multi-tenant architecture, and applied LLM pipelines.

Experience and Projects

Siemens – *Technical Intern, Software and Logic Automation*

June 2025 – August 2025

- Killed a class of undebuggable runtime failures – modelled a real-time closed-loop control system as a **two-interrupt state machine** managing 4 concurrent actuator transitions with zero runtime debugger.
- Made execution flow across **3 SCADA workflows** fully auditable at factory scale – modelled as explicit state machines in Siemens PLC/HMI; zero ambiguity in physical actuator states.

LinkedOut – **Multi-Tenant AI Outreach System**

[GitHub](#)

FastAPI, PostgreSQL, pgvector, AWS EC2/S3, Redis, OAuth2/JWT, asyncio, Docker, Next.js, Playwright

- Closed **100% of cross-user data exposure** – traced live data-leak to shared SQLAlchemy session state; fixed with per-request session isolation via FastAPI dependency injection.
- Drove **EC2 IP ban rate from 30% to 0%** – traced silent bans (zero results, zero error logs) to browser fingerprinting; fixed with Playwright stealth mode, randomised delays, and cookie reuse.
- Engineered **full multi-tenant isolation** – async FastAPI with ThreadPoolExecutor for browser task isolation, OAuth2/JWT auth, **user_id**-scoped DB tables, per-user S3 namespacing, 15 req/day rate limiting.
- Removed manual pipeline re-initialisation entirely – wired **LangGraph human-approval checkpointing** via `interrupt()` and `MemorySaver`; pipelines pause, wait for review, and resume with full state intact.

CargoNova AI – **Shipment Validation Pipeline**

[GitHub](#)

LangGraph, Groq (Llama 4 Scout), FastAPI, Streamlit, SQLite, Gmail SMTP, TF-IDF, pypdf, Watchdog

- Replaced a fully manual 2–4 loop-per-shipment process – shipped an **end-to-end AI validation pipeline** that extracts 8 key fields from invoice/BOL/packing list via Llama 4 Scout and cross-validates consistency across docs.
- Designed to handle messy real-world docs – port aliases, Incoterm variants, and naming inconsistencies resolved correctly via **fuzzy matching**; no brittle exact-match rules.
- Designed **three-tier decision engine** (`auto_approve` / `flag_for_review` / `amendment_required`) with LLM-drafted reply emails; human-in-the-loop by design – cargo team reviews and clicks Approve & Send.
- Zero-click pipeline trigger – **Watchdog + Gmail listener** fires the full pipeline on new shipments automatically; all results persisted to SQLite with a plain-English audit query interface.

VaultAI – **Multimodal Document Q&A System**

[GitHub](#) [Live Demo](#)

FastAPI, FAISS, HuggingFace Embeddings, PostgreSQL, OpenAI Whisper, LangChain, FFmpeg, AWS EC2, React

- Enabled **timestamp-linked AI playback** from any answer – video stripped to audio via FFmpeg, transcribed segment-by-segment with Whisper, each segment stored in PostgreSQL with start timestamp and ingested into FAISS independently.
- Zeroed out hallucinations by design – **retrieval-grounded LangChain + Groq LLaMA 3.3** returns top-3 L2-scored chunks; strict prompt forces "I don't know" when no relevant chunk found, never fabricates.
- Full source traceability per response – **content-type-aware chunking** with `chunk_id` metadata at ingest: `RecursiveCharacterTextSplitter` for documents, raw Whisper segments for audio/video.

Technical Skills

Languages: Python, C++, JavaScript, SQL

Backend and APIs: FastAPI, Flask, Node.js, REST API Design, Microservices, asyncio, WebSockets, OAuth2, JWT, Rate Limiting

Databases and Storage: PostgreSQL, pgvector, Redis, FAISS, Supabase, AWS S3, SQLite

Cloud and DevOps: AWS (EC2, S3), Docker, Nginx, Certbot (SSL/TLS), CI/CD, GitHub Actions

System Design: Multi-tenant Architecture, Concurrency and Session Isolation, Caching Strategies, Async Task Queues, API Design

AI/ML (Applied): PyTorch, Hugging Face, Sentence Transformers, LLM Inference, Vector Embeddings, RAG, LangGraph

Frontend and Tools: React, Next.js, Tailwind CSS, Playwright, Git, Postman, Streamlit

Achievements

- **Global Rank ~4,000** out of 30,000+ in LeetCode Biweekly Contest – top 13%.
- Solved **200+ DSA problems** across LeetCode, CodeStudio, and GeeksforGeeks.

Education

Dr. B. R. Ambedkar National Institute of Technology (NIT) Jalandhar

July 2022 – June 2026

Bachelor of Technology (B.Tech)